

Lecture 12

Introduction to Computational Quantum Chemistry: Modeling electronic properties

Source: lectures by Dr. L Slipchenko

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Energy scales

Objects of quantum mechanics (QM): atoms, electrons and interactions between them:

- Electron ionization, electron excitation, molecular dissociation and chemical reactions (0.001 – 0.1 Hartree) (30 meV – 3 eV)
- Molecular vibrations ($10^{-3} - 10^{-2}$ H) (30 meV – 300 meV)
- Molecular rotations ($10^{-6} - 10^{-4}$ H) (30 μ eV – 3 meV)
- Interactions of electron and nuclear spins ($< 10^{-4}$ H) (<3 meV)

Room temp:
 $kT \sim 25$ meV
 $\sim 10^{-3}$ H

Energy of H atom: -0.5 Hartree, ~ -13 eV
Energy of ethylene C_2H_4 : ~ 80 Hartree, ~ 3 keV

1 Hartree: atomic units, where electron mass and charge, Planck $\hbar$, and permittivity of free space ϵ_0 all are =1.

1 Hartree (H) = 1 a.u. =
27.212 eV =
627.51 kcal/mol =
 4.36×10^{-18} J =
 2.195×10^5 cm $^{-1}$ =
 3.158×10^5 K

We need to capture relatively small changes of total molecular energies:

Electronic structure calculations need to be conducted with extremely high level of accuracy!

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Molecular simulations

Electronic structure theory / Quantum Chemistry

Solves for motion of electrons in molecules; provides static properties of individual molecules and potential energy surfaces

Molecular dynamics

Solve for motions of molecules on a (quantum or classical) potential energy surface; provides reaction rates

Statistical Mechanics

Solve for bulk properties from properties of individual molecules

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Quantum and classical molecular simulations

Molecules consist of nuclei & electrons ->
Laws describing interactions of electrons ->
Quantum mechanics (QM)

Electrons are represented by wave function ->
Schrodinger eqn (**SE**) (1925)

Electronic motion
is parameterized :
Force fields (FF)

Molecules consist of atoms (classical particles) ->
Laws describing motion of classical particles ->
Classical or molecular mechanics (MM)

Newton's laws of motion (1687)

Quantum	Classical
More accurate	Less accurate
Equations are more complicated	Equations are less complicated
Computationally very expensive (typically cannot do dynamics)	Computationally less expensive (can do dynamics)
Can describe the motion of electrons: UV/vis spectra, electron transfer, etc.	Cannot describe the motion of electrons
No empirical parameters or only a few (e.g. DFT)	Involves many empirical parameters

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What can we learn from solving SE?

- Geometrical structures (rotational spectra)
- Rovibrational energy levels → infrared and Raman spectra
- Potential energy surfaces (barrier heights, transition states); with a treatment of dynamics → reaction rates and mechanisms
- Quantum Mechanics + Statistical Mechanics → Thermochemistry (ΔS , ΔH , ΔG , C_v , C_p)
- Electronic energy levels (UV and visible spectra)
- Ionization potentials (photoelectron and X-ray spectra)
- Electron affinities
- Franck-Condon factors (transition probabilities, vibronic intensities)
- Dipole moments, polarizabilities
- Electron density maps and population analyses
- Magnetic shielding tensors → NMR spectra

All structural and energetic information
on a molecular system

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Good news / bad news

“The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble.”

P.A.M. Dirac,
Proc. Roy. Soc. (London), 123 714 (1929)



→ **NEED APPROXIMATIONS**

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Numerical solution of SE: art of approximations

Approximations:

- Born-Oppenheimer approximation
- Representation of wave function in a basis set
- Approximate treatment of electron correlation (interactions between electrons)

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Molecular non-relativistic Hamiltonian in atomic units

$$\hat{H}\Psi = E\Psi$$

$$\hat{H} = \hat{T}_N(R) + \hat{T}_e(r) + \hat{V}_{NN}(R) + \hat{V}_{eN}(r, R) + \hat{V}_{ee}(r)$$

$$\hat{T}_N = -\sum_i \frac{1}{2M_A} \nabla_A^2$$

kinetic energy of nuclei

$$\hat{T}_e = -\frac{1}{2} \sum_i \nabla_i^2$$

kinetic energy of electrons

$$\hat{V}_{NN} = \sum_{A>B} \frac{Z_A Z_B}{R_{AB}}$$

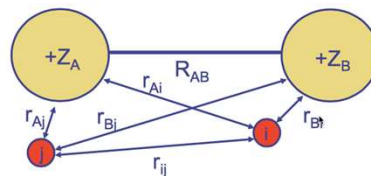
potential energy of nuclei

$$\hat{V}_{eN} = -\sum_{A,i} \frac{Z_A}{r_{Ai}}$$

potential energy of interactions of nuclei and electrons

$$\hat{V}_{ee} = \sum_{i>j} \frac{1}{r_{ij}}$$

potential energy of electrons



atomic units
 $\hbar = 1$
 $e = 1$
 $a_0 = 1$
 $m_e = 1$

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Born-Oppenheimer approximation

$$\hat{H}(\mathbf{r}, \mathbf{R}) = \hat{T}_N(\mathbf{R}) + \hat{T}_e(\mathbf{r}) + \hat{V}_{NN}(\mathbf{R}) + \hat{V}_{eN}(\mathbf{r}, \mathbf{R}) + \hat{V}_{ee}(\mathbf{r})$$

In non-relativistic SE, the electronic and nuclear degrees of freedom are coupled:

$$\hat{V}_{eN}(\mathbf{r}, \mathbf{R}) = -\sum_{A,i} \frac{Z_A}{r_{Ai}}$$

$\hat{V}_{eN}(\mathbf{r}, \mathbf{R})$ - large, cannot be neglected

The Born-Oppenheimer approximation:
the separation into nuclear and electronic coordinates is approximately correct

The nuclei are much more massive than the electrons $\rightarrow$ the nuclei are nearly fixed wrt electron motion $\rightarrow$ assume that dependence on $\mathbf{R}$ is **parametric**: $\Psi(\mathbf{r}, \mathbf{R}) = \Phi_e(\mathbf{r}, \{\mathbf{R}\}) \Phi_N(\mathbf{R})$



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Born-Oppenheimer approximation

$T_N \ll T_e$ (by a factor of M_A/μ_e) $\rightarrow T_N \sim 0$

Electronic Hamiltonian for **fixed** nuclear coordinates:

$$\hat{H}_{el} = \hat{T}_e(\mathbf{r}) + \underbrace{\hat{V}_{NN}(\mathbf{R})}_{\text{constant}} + \underbrace{\hat{V}_{eN}(\mathbf{r}; \mathbf{R})}_{\text{parametrically depend on } \mathbf{R}} + \hat{V}_{ee}(\mathbf{r})$$

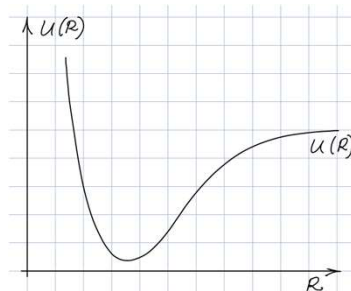
Solve "clamped-nuclei" SE for a given value of $\mathbf{R}=\mathbf{R}_a$:

$$\hat{H}_{el}\Phi_e(\mathbf{r}; \{\mathbf{R}_a\}) = U(\mathbf{R}_a)\Phi_e(\mathbf{r}; \{\mathbf{R}_a\})$$

Repeat such calculations for different values of $\mathbf{R}_a \rightarrow$ obtain molecular potential energy function:

$$U(\mathbf{R}) = E_e(\mathbf{R}) + \sum_{A>B} \frac{Z_A Z_B}{R_{AB}}$$

This is essence of **electronic structure** or **ab initio** calculations



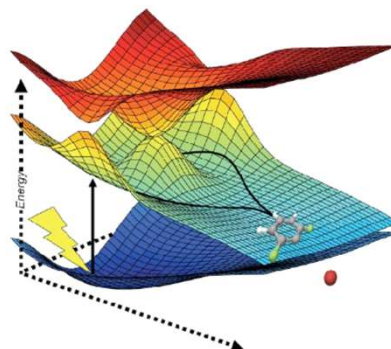
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Potential energy surfaces: polyatomic molecules

$$U_k(\mathbf{R}) = E_{e,k}(\mathbf{R}) + \sum_{A>B}^{nuc} \frac{Z_A Z_B}{R_{AB}}$$

$U_k(\mathbf{R})$ are potential energy surfaces for nuclei, imposed by electrons in excitation state k

- Potential energy surface for a polyatomic molecule is generally very complicated
- Different solutions of electronic SE give different (excited state) surfaces
- A molecule might undergo electronic transitions (electronic excitations) upon interactions with light, electromagnetic field, mechanical stress, high temperature, etc.
- Electronic structure calculations: solve for electronic Schrödinger equation, get electronic properties and potential energy surfaces
- Dynamics: solve for the motion of nuclei on the potential energy surfaces. Important for a detailed understanding of reaction mechanisms



$\mathbf{R}$ is $(3N-6)$ -dimensional for non-linear N -atomic molecules (linear $3N-5$)

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Electronic wave function

electronic coordinates: $\mathbf{x} = \{r, \omega\}$ ← Coordinate (r) and spin (ω)

$\alpha(\omega)$ – spin up, $\beta(\omega)$ – spin down

$\mathbf{x} \rightarrow (x, y, z, \omega)$

spin orbital $\chi(\mathbf{x})$ – describes both spatial and spin coordinates $\chi(\mathbf{x}) = \begin{cases} \psi(r)\alpha(\omega) \\ \psi(r)\beta(\omega) \end{cases}$

In general N -electron case: **Slater determinant** (see Lecture 1-5)

$$\Psi(x_1, x_2, \dots, x_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_i(x_1) & \chi_j(x_1) & \dots & \chi_k(x_1) \\ \chi_i(x_2) & \chi_j(x_2) & \dots & \chi_k(x_2) \\ \dots & \dots & \dots & \dots \\ \chi_i(x_N) & \chi_j(x_N) & \dots & \chi_k(x_N) \end{vmatrix}$$

electrons

$$= |\chi_i(x_1)\chi_j(x_2)\dots\chi_k(x_N)\rangle$$

spin orbitals

Note: for each term:

$$\hat{H}\Psi^{HP} = E\Psi^{HP}$$

$$\Psi^{HP}(x_1, x_2, \dots, x_N) = \chi_i(x_1)\chi_j(x_2)\dots\chi_k(x_N) \quad \text{Hartree product wave function}$$

$$E = \varepsilon_i + \varepsilon_j + \dots + \varepsilon_k \quad \text{energies of spin-orbitals } i, j, k \dots \text{ occupied by electrons } 1, 2, 3 \dots$$

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Variational principle

Variational Theorem:

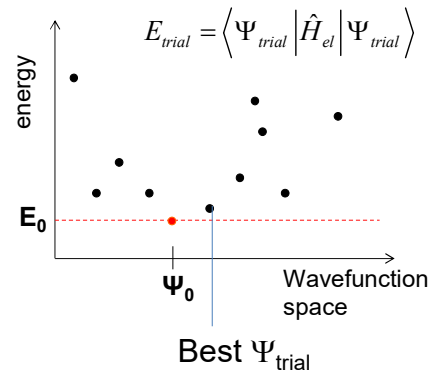
the exact ground state energy of any quantum system E_0 is invariably smaller than the approximate ground state energy E_0' obtained using *any* approximate trial ground state wavefunction Ψ_0' .

$$E_0 \leq E_0'$$

Lower energy $\leftrightarrow$ better (closer to the exact eigenvalue) wavefunction

The bigger the allowed wavefunction subspace $\rightarrow$ lower energy can be found

Variational theorem allows us to improve the wavefunction by trying to find one that results in the lowest possible Hamiltonian expectation value (ground state energy).



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Hartree-Fock model

The Hartree-Fock model tries to find the **best variational solution** to the electronic Hamiltonian, assuming that the wavefunction is approximated by a **single Slater determinant**.

This is the same thing as assuming that **each electron only feels an average charge distribution due to other electrons** (mean field)



Douglas
Hartree

1897-1958



Vladimir Fock

1898-1974

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Hartree-Fock (HF) model

Hartree-Fock model tries to find the *best variational solution* to the electronic Hamiltonian assuming that the wave function is approximated by a **single Slater determinant**.

General strategy:

1. $E_{HF} = \langle \Psi_{SD} | \hat{H}_{el} | \Psi_{SD} \rangle$ - this equation defines HF energy as a functional of spin-orbitals
2. Minimize E_{HF} with respect to spin-orbitals: $\delta E_{HF}[\{\chi_i\}] = 0 \rightarrow$
this equation determines optimal spin-orbitals and the variationally best (lowest) energy
3. Solve this equation for the best orbitals and energy – job done!

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Notations for electronic Hamiltonian

$$\text{Electronic Hamiltonian } \hat{H}_{el} = - \sum_i^{elec} \frac{1}{2} \nabla_i^2 - \sum_i^{elec} \sum_A^{nuc} \frac{Z_A}{r_{Ai}} + \sum_{i < j}^{elec} \frac{1}{r_{ij}} + \hat{V}_{NN}$$

$$\hat{h}(i) = -\frac{1}{2} \nabla_i^2 - \sum_A \frac{Z_A}{r_{iA}}$$

one-electron operator:

accounts for kinetic energy and Coulomb attraction to all nuclei for electron i

$$\hat{v}(i, j) = \frac{1}{r_{ij}}$$

two-electron operator:

Coulomb repulsion between electrons i and j

$$\hat{H}_e = \sum_i \hat{h}(i) + \sum_{i < j} \hat{v}(i, j) + \hat{V}_{nn}$$

electronic Hamiltonian in these operators

↑
Potential energy of nuclei

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Hartree-Fock (HF) solutions

$$\hat{H}_e = \sum_i \hat{h}(i) + \sum_{i<j} \hat{v}(i,j) + \hat{V}_{nn} \quad \hat{h}(i) = -\frac{1}{2} \nabla_i^2 - \sum_A \frac{Z_A}{r_{iA}} \quad \hat{v}(i,j) = \frac{1}{r_{ij}}$$

Electronic Hamiltonian one-electron operator two-electron operator

Expectation value of energy corresponding to a Slater determinant (Ψ_{SD}) wf:

$$E_{HF} = \langle \Psi_{SD} | \hat{H}_e | \Psi_{SD} \rangle = \sum_i^{elec} \langle i | \hat{h} | i \rangle + \sum_{i>j}^{elec} (\langle ij | ij \rangle - \langle ij | ji \rangle)$$

↑ Like coulomb integral ↑ exchange integral

one-electron integral (4-dimensional, [coordinate, spin]) Slang, it is

$$\langle i | \hat{h} | j \rangle = \int dx_1 \chi_i^*(x_1) \hat{h}(r_1) \chi_j(x_1) \quad \left\langle \chi_i(x_1) \chi_j(x_2) \left| \frac{1}{r_{12}} \right| \chi_j(x_1) \chi_i(x_2) \right\rangle$$

two-electron integral (8-dimensional)

$$\langle ij | kl \rangle = \int dx_1 dx_2 \chi_i^*(x_1) \chi_j^*(x_2) \frac{1}{r_{12}} \chi_k(x_1) \chi_l(x_2)$$

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Physical meaning of HF solution

$$\hat{H}_e = \sum_i \hat{h}(i) + \sum_{i<j} \hat{v}(i,j) + \hat{V}_{nn} \quad \hat{h}(i) = -\frac{1}{2} \nabla_i^2 - \sum_A \frac{Z_A}{r_{iA}} \quad \hat{v}(i,j) = \frac{1}{r_{ij}}$$

Electronic Hamiltonian one-electron operator two-electron operator

HF energy: $E_{HF} = \langle \Psi_{SD} | \hat{H}_e | \Psi_{SD} \rangle = \sum_i^{elec} \langle i | \hat{h} | i \rangle + \sum_{i>j}^{elec} (\langle ij | ij \rangle - \langle ij | ji \rangle)$

Each electron contributes a one-electron integral:

$$\langle i | \hat{h} | i \rangle = \int dx_1 \chi_i^*(x_1) \hat{h}(r_1) \chi_i(x_1)$$

This looks like the expectation (average) value of the operator $\hat{h}$ for an electron in orbital χ_i

Recall: $\hat{h}$ contains electron kinetic energy and potential of attraction to all nuclei

Sum over all (occupied by electrons) orbitals i gets the total kinetic energy and attraction to nuclei

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Physical meaning of HF solution

$$\hat{H}_e = \sum_i \hat{h}(i) + \sum_{i<j} \hat{v}(i,j) + \hat{V}_{nn} \quad \hat{h}(i) = -\frac{1}{2} \nabla_i^2 - \sum_A \frac{Z_A}{r_{iA}} \quad \hat{v}(i,j) = \frac{1}{r_{ij}}$$

Electronic Hamiltonian one-electron operator two-electron operator

HF energy: $E_{HF} = \langle \Psi_{SD} | \hat{H}_e | \Psi_{SD} \rangle = \sum_i^{elec} \langle i | \hat{h} | i \rangle + \sum_{i>j}^{elec} (\langle ij | ij \rangle - \langle ij | ji \rangle)$

Each pair of electrons (in orbitals i and j) has a Coulomb integral:

$$\langle ij | ij \rangle = \int dx_1 dx_2 \chi_i^*(x_1) \chi_j^*(x_2) \frac{1}{r_{12}} \chi_i(x_1) \chi_j(x_2)$$

Integrate over all possible locations for the electrons

Probability electron 1 in orbital i is located at x_1

Probability electron 2 in orbital j is located at x_2

Coulomb repulsion between electron at x_1 and electron at x_2

Overall, this integral represents the Coulomb repulsion between electron 1 in orbital i and electron 2 in orbital j

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Physical meaning of HF solution

$$\hat{H}_e = \sum_i \hat{h}(i) + \sum_{i<j} \hat{v}(i,j) + \hat{V}_{nn} \quad \hat{h}(i) = -\frac{1}{2} \nabla_i^2 - \sum_A \frac{Z_A}{r_{iA}} \quad \hat{v}(i,j) = \frac{1}{r_{ij}}$$

Electronic Hamiltonian one-electron operator two-electron operator

HF energy: $E_{HF} = \langle \Psi_{SD} | \hat{H}_e | \Psi_{SD} \rangle = \sum_i^{elec} \langle i | \hat{h} | i \rangle + \sum_{i>j}^{elec} (\langle ij | ij \rangle - \langle ij | ji \rangle)$

Each pair of electrons (in orbitals i and j) also has an *exchange* integral:

$$\langle ij | ji \rangle = \int dx_1 dx_2 \chi_i^*(x_1) \chi_j^*(x_2) \frac{1}{r_{12}} \chi_j(x_1) \chi_i(x_2)$$

Somewhat similar to exchange term in Dexter energy transfer)

$$\langle ij | ji \rangle = \int dx_1 dx_2 \chi_i^*(x_1) \chi_j(x_1) \frac{1}{r_{12}} \chi_j^*(x_2) \chi_i(x_2)$$

Overlap of electron on orbitals i and j

No classical analogy... Physical meaning: an electron repels another electron of the same spin (so called Fermi hole). This integral is a consequence of the Slater determinant (antisymmetrized) wavefunction

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Variational principle for HF model

$$\hat{H}_e = \sum_i \hat{h}(i) + \sum_{i<j} \hat{v}(i,j) + \hat{V}_{nn} \quad \hat{h}(i) = -\frac{1}{2} \nabla_i^2 - \sum_A \frac{Z_A}{r_{iA}} \quad \hat{v}(i,j) = \frac{1}{r_{ij}}$$

Electronic Hamiltonian one-electron operator two-electron operator

$$\text{HF energy: } E_{HF} = \langle \Psi_{SD} | \hat{H}_e | \Psi_{SD} \rangle = \sum_i^{elec} \langle i | \hat{h} | i \rangle + \sum_{i>j}^{elec} (\langle ij | ij \rangle - \langle ij | ji \rangle)$$

HF energy depends on spin-orbitals (electron wave functions).

How to find them? -> **Variational principle!**

Find minimum of energy by varying spin-orbitals: $\delta E_{HF} \left[\{ \chi_i \} \right] = 0$

This results in the following set of equations on spin-orbitals:

HF eigenvalue problem

$$\hat{f}(x_1) \chi_i(x_1) = \varepsilon_i \chi_i(x_1)$$

Fock operator

$$\hat{f}(x_1) = \hat{h}(x_1) + \sum_j (\hat{J}_j(x_1) - \hat{K}_j(x_1))$$

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HF eigenvalue equations

HF eigenvalue problem

$$\hat{f}(x_1) \chi_i(x_1) = \varepsilon_i \chi_i(x_1)$$

Fock operator

$$\hat{f}(x_1) = \hat{h}(x_1) + \sum_j (\hat{J}_j(x_1) - \hat{K}_j(x_1))$$

Fock operator has a functional dependence through Coulomb (J) and exchange (K) operators on solution $\{\chi\}$

HF equations are non-linear and should be solved iteratively until self-consistency

$$\hat{J}_j(x_1) \chi_i(x_1) = \left[\int dx_2 \frac{\chi_j^*(x_2) \chi_j(x_2)}{r_{12}} \right] \chi_i(x_1)$$

$$\hat{K}_j(x_1) \chi_i(x_1) = \left[\int dx_2 \frac{\chi_j^*(x_2) \chi_i(x_2)}{r_{12}} \right] \chi_j(x_1)$$

Hartree-Fock (HF) $\equiv$
Self Consistent Field (SCF)

Hartree-Fock procedure:

1. Make an initial guess of 2K spin-orbitals $\{\chi_i^{(0)}\}$
2. Calculate Fock operator for each spin-orbital $\hat{f}_i = \hat{h}_i + \sum_j^{Nelec} (\hat{J}_j - \hat{K}_j)$
3. Solve $\hat{f}_i |\chi_i\rangle = \varepsilon_i |\chi_i\rangle$, obtain a new set of spin-orbitals $\{\chi_i^{(1)}\}$
4. **Repeat until self-consistency**, i.e. until $\{\chi_i^{(k+1)}\} = \{\chi_i^{(k)}\}$

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Introducing basis set

- Approximation: One-electron wave functions (orbitals) can be expanded in the basis set:

$$\chi_i = \sum_{\mu=1}^K C_{\mu i} \phi_{\mu} \quad \text{basis } \phi_{\mu}, \mu=1 \dots K$$

- HF equations become matrix equations: perfect for solving on a computer!

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Basis sets

The spin orbitals are expressed as a superposition of a set of (known) basis functions:

$$\chi_i = \sum_{\mu=1}^K C_{\mu i} \phi_{\mu} \quad \text{basis } \phi_{\mu}, \mu=1 \dots K$$

HF eigenvalue problem

$$\hat{f}(x_1) \chi_i(x_1) = \varepsilon_i \chi_i(x_1)$$

$$\hat{f}(x_1) = \hat{h}(x_1) + \sum_j (\hat{J}_j(x_1) - \hat{K}_j(x_1))$$

$$\hat{f} \sum_{\nu} C_{\nu i} \phi_{\nu} = \varepsilon_i \sum_{\nu} C_{\nu i} \phi_{\nu}$$

$$\langle \phi_{\mu} | \times$$

$$\sum_{\nu} C_{\nu i} \int dx_1 \phi_{\mu}^*(x_1) \hat{f}(x_1) \phi_{\nu}(x_1) = \varepsilon_i \sum_{\nu} C_{\nu i} \int dx_1 \phi_{\mu}^*(x_1) \phi_{\nu}(x_1)$$

$F_{\mu\nu}$
Fock matrix

$S_{\mu\nu}$
overlap matrix

$$\sum_{\nu} F_{\mu\nu} C_{\nu i} = \varepsilon_i \sum_{\nu} S_{\mu\nu} C_{\nu i} \quad \text{Roothaan Equations}$$

$$\mathbf{FC} = \mathbf{SC}\boldsymbol{\varepsilon} \quad (\text{matrix form})$$



Clemens Roothaan
(1918 – 2019)

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HF self-consistent field procedure

1. Specify molecular geometry, basis functions, and electronic state of interest (i.e., singlet, triplet, etc)
2. Form overlap matrix **S**
3. Guess the initial molecular orbital coefficients **C**
4. Form Fock matrix **F**
5. Solve $\mathbf{FC} = \mathbf{SC}\epsilon$
6. Use new molecular orbital coefficients **C** to build a new Fock matrix **F**
7. Go to step 5; repeat until **C** no longer changes from one iteration to the next

The HF method boils down to the matrix diagonalization problem.

But the Fock matrix **F** depends on its eigenvectors **C**, which is why we need iterations to bring **F** and **C** into consistency

Computational cost of the HF method:

- computing the Fock matrix involves computing $\sim K^4$ 2-electron integrals (K is the number of basis functions)
 - diagonalization of the Fock matrix of dimensionality $K \times K$ is proportional to K^3
- cost $\sim K^3 - K^4$ depending on implementation

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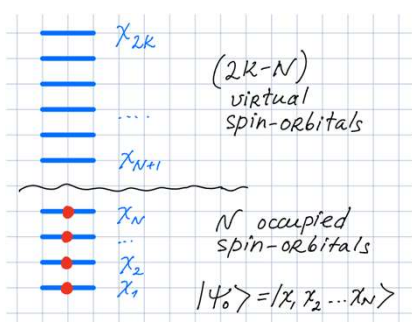
Hartree-Fock solutions

After solving $2K$ HF eq-ns:

$$\hat{f}_i |x_i\rangle = \epsilon_i |x_i\rangle$$

K – the total number of spatial orbitals,
determined by the total number of basis functions
 $2K$ – the total number of spin-orbitals

$2K$ orthonormal spin-orbitals $\{x_i\}$ with orbital energies $\{\epsilon_i\}$ is produced



The Slater determinant formed from the N lowest occupied spin-orbitals $|\Phi_0\rangle = |x_1 x_2 \dots x_N\rangle$ is the Hartree-Fock ground state wave function.

It is the best variational approximation to the ground state of the system in the single determinantal form.

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Hartree-Fock solutions

$\hat{f}_i |\chi_i\rangle = \varepsilon_i |\chi_i\rangle$ 2K orthonormal spin-orbitals $\{\chi_i\}$ with orbital energies $\{\varepsilon_i\}$ is produced

Q: Is the Hartree-Fock ground state wave function the exact solution to the system?

Q: How to improve it?

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The exact solution to N-electron problem

Given N electrons and 2K spin orbitals

The best solution is to combine together **all possible valid determinants** $\Psi = \sum_i c_i \Phi_i$ and use the variational principle to minimize the ground state energy.

This model is called **Full Configuration Interaction (FCI)**

FCI is the exact solution of electronic SE in a given spin-orbital basis (with given K)

How big is a FCI wave function? The number of possible SDs is given as the number of times we can draw N spin-orbitals (N electrons) from a pool of 2K:

$$\# \text{configurations} = C_{2K}^N = \frac{(2K)!}{N!(2K-N)!} \approx 10^N$$

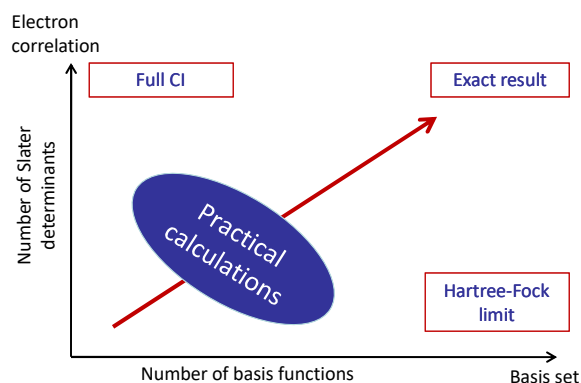
$$\# \text{ config} = C_{2K}^N = \frac{(2K)!}{N!(2K-N)!}$$

N	2K	C_{2K}^N
2	10	45
4	20	4845
6	30	593775
8	40	$7.7 \cdot 10^7$
10	50	$1.0 \cdot 10^{10}$
20	100	$5.4 \cdot 10^{20}$

Today largest computations can deal with $\sim 10^9$ configurations

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Theoretical model chemistries (John Pople)



Two approximations:

- We do not solve the exact equations → approximate treatment of electron-electron interactions (aka electron correlation)
- We do not achieve completeness in the electron expansion space (we do not solve inexact equations exactly) → basis set truncation error

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Choosing the basis set

- Basis functions should have a behavior that agrees with the physics of the problem
- Basis functions should make it easy to calculate all the required integrals

Molecules

- Gaussian functions (integrals are semi-analytical)
- exponential functions (close resemblance to H atom eigenfunctions)

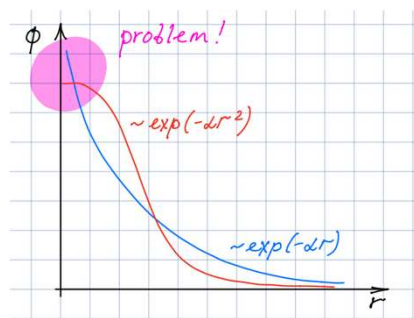
Crystals/solids

- plane waves

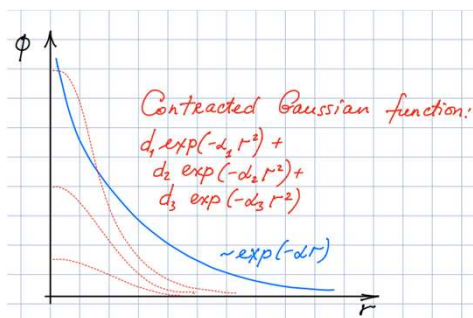
A more complete (larger) basis set will result in a *variationally* better solution (lower HF energy) – but not necessarily a practically better solution!

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Gaussian basis sets: contractions



Gaussian functions are not perfect



Solution: **Contracted Gaussian functions**

Choose in advance the number of Gaussians, their widths and weights → this composition becomes a single predefined basis function

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Common Gaussian basis sets

Minimal basis set (**STO-3G**)

H, He: 1 (1s)

Li ... Ne: 5 (1s2s2p_{xyz})

Na ... Ar: 9 (1s2s2p_{xyz}3s3p_{xyz})

STO-3G = Slater-Type Orbital approximated by 3 Gaussians
Optimized for each atom separately

How they are constructed:

- Use a very large basis, solve variationally, and find the 3G best fit

too small for chemical accuracy

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Common Gaussian basis sets

Let's allow some expansion, contraction and anisotropy of orbitals

Double zeta (DZ orbitals):

The Greek ζ (zeta) represents the exponent in orbital $e^{-\zeta r}$

- Uses 2 b.f. for each atomic orbital, : "tight" and "diffuse",
- Two b.f. allow the reshaping of the size of the orbital

e.g.: e.g. C: 1s, 1s', 2s, 2s', 2p, 2p'

check for **DZ**, e.g. **DZP**, cc-p**VDZ**

Tripple zeta (3 b.f. for each atomic orbital): "tight", "medium", "diffuse."

e.g. C: 1s, 1s', 1s'', 2s, 2s', 2s'', 2p, 2p', 2p''

check for **TZ**, e.g. **TZP**, aug-cc-p**VTZ**

quadruple zeta (4 b.f. for each atomic orbital): **QZ**

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Common Gaussian basis sets

Split-valence bases (J. Pople):

6-31G split-valence double-zeta

inert core, valence orbitals e.g. C: **1s**, **2s**, **2s'**, **2p**, **2p'**

1 b.f. for all core electrons (e.g., 1s) constructed as superposition of **6** Gaussians

2 b.f. for valence electrons (double-zeta) contracted from 3 and 1 Gaussians

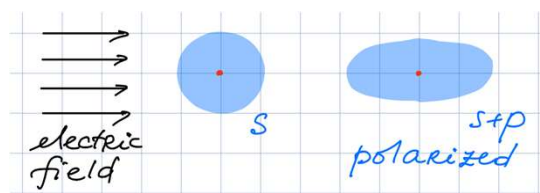
6-31G

Uses Gaussians

dash = split-valence

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Common Gaussian basis sets



Polarization functions: higher angular momentum functions that allow anisotropy

- Allows orbitals to tilt/bend toward other atoms
- more important for heavy elements than for Hydrogen
- important for ions, polar molecules, molecules with electronegative atoms (F, O, etc)

Check for letter “p” in the basis name: DZ**P**, cc-pVDZ

Pople bases: * or explicit names of added functions: 6-31G** = 6-31G(d,p)

non-Hydrogens
D-type functions

Hydrogens
p-type functions

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Common Gaussian basis sets

Diffuse functions: functions with very shallow Gaussians, can describe electron density far from nuclei

- more important for heavy elements than for Hydrogen
- important for anions, electronic excited states, non-covalent complexes

Check for letter “+” or “aug” in the basis name: DZP+, aug-cc-pVDZ, 6-31+G*

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Common Gaussian basis sets

description	Pople	Dunning	Correlation-consistent
Double-zeta polarized	6-31G* 6-31G**	DZP	cc-pVDZ
Double-zeta polarized with diffuse functions	6-31+G* 6-31++G**	DZP+	aug-cc-pVDZ
Triple-zeta polarized	6-311G* 6-311G**	TZP	cc-pVTZ
Triple-zeta polarized with diffuse functions	6-311+G* 6-311++G**	TZP+	aug-cc-pVTZ
Quadruple-zeta polarized	-	-	cc-pVQZ
...			

See more: [https://en.wikipedia.org/wiki/Basis_set_\(chemistry\)](https://en.wikipedia.org/wiki/Basis_set_(chemistry))